No.

Date

1.

t (s)	0	1	2	3	4	5	6
$F(t)$	0.0	3.2	5.1	6.0	5.4	3.1	0.4

Trapesium

$a = 0$, $b = 6$, $n = 6$, $h = 1$

$$J_T \approx \frac{h}{2}\left[F_0 + 2F_1 + 2F_2 + 2F_3 + 2F_4 + 2F_5 + F_6\right]$$

$$\approx \frac{1}{2}\left[0.0 + 2(3.2) + 2(5.1) + 2(6.0) + 2(5.4) + 2(3.1) + 0.4\right]$$

$$\approx \frac{1}{2}\left[0.0 + 6.4 + 10.2 + 12.0 + 10.8 + 6.2 + 0.4\right]$$

$$\approx \frac{1}{2}(46.0) = 23 \text{ kN.s} = 23000 \text{ N.s}$$

Simpson 1/3

$$J_S \approx \frac{h}{3}\left[F_0 + 4F_1 + 2F_2 + 4F_3 + 2F_4 + 4F_5 + F_6\right]$$

$$\approx \frac{1}{3}\left[0.0 + 4(3.2) + 2(5.1) + 4(6.0) + 2(5.4) + 4(3.1) + 0.4\right]$$

$$\approx \frac{1}{3}\left[0.0 + 12.8 + 10.2 + 24.0 + 10.8 + 12.4 + 0.4\right]$$

$$\approx \frac{1}{3}(70,6) = 23,5333 \text{ kN.s} = 23533,3 \text{ N.s}$$

$$\Delta V_T = \frac{J}{m} = \frac{23000}{50} = 460 \text{ m/s}$$

$$\Delta V_S = \frac{J}{m} = \frac{23533,3}{50} = 470,67 \text{ m/s}$$

$$\text{selisih} = \frac{J_S - J_T}{J_S} \times 100\% = \frac{23533,3 - 23000}{23533,3} \times 100\% = \frac{0,5333}{23533,3} \times 100\% = 2,2663\%$$

2.

X (mm)	-0,3	-0,2	-0.1	0.0	0.1	0.2	0.3
U (μJ)	0.500	0,220	0.055	0.000	0.060	0.250	0.590

selisih pusat du/dx di $x = 0.0$

$$\left.\frac{du}{dx}\right|_{x=0} \approx \frac{U(0.1) - U(-0.1)}{2h} = \frac{0.060 - 0.055}{2(0.1)} = \frac{0.005}{0.2} = 0.0250 \ \mu\text{J/mm}$$

Gaya $F(0.0) \rightarrow F(0.0) = -\left.\frac{du}{dx}\right|_{x=0} = -0.0250$ mN selisih pusat $\frac{d^2u}{dx^2}$ di $x = 0.0$

$$\left.\frac{d^2u}{dx^2}\right|_{x=0} \approx \frac{u(0.1) - 2u(0.0) + u(-0.1)}{h^2}$$

$$= \frac{0.060 - 2(0.000) + 0.055}{(0.1)^2} = \frac{0.115}{0.01} = 11.5 \ \mu J/mm^2$$

Selisih maju di $x = -0.3$ mm

$$\left.\frac{du}{dx}\right|_{x=-0.3} \approx \frac{u(-0.2) - u(-0.3)}{h} = \frac{0.220 - 0.500}{0.1} = -2.8 \ \mu J/mm$$

$$\left.\frac{d^2u}{dx^2}\right|_{x=-0.3} = \frac{u(-0.1) - 2u(-0.2) + u(-0.3)}{h^2} = \frac{0.055 - 2(0.220) + 0.500}{0.01}$$

$$= \frac{0.115}{0.01} = 11.5 \ \mu J$$

Selisih mundur di $x = 0.3$ mm

$$\left.\frac{du}{dx}\right|_{x=0.3} \approx \frac{u(0.3) - u(0.2)}{h} = \frac{0.590 - 0.250}{0.1} = 3.4 \ \mu J/mm$$

$$\left.\frac{d^2u}{dx}\right|_{x=0.3} \approx \frac{u(0.3) - 2u(0.2) + u(0.1)}{h^2} = \frac{0.590 - 2(0.250) + 0.060}{0.01}$$

$$= \frac{0.150}{0.01} = 15 \ \mu J/mm^2$$

Karena $d^2u/dx^2|_{x=0} = 11.5 > 0$, energi potensial berbentuk cekung ke atas di sekitar $x = 0$. maka $x = 0.0$ merupakan titik kesetimbangan stabil karena setiap penyimpangan kecil dari titik ini akan menghasilkan gaya pemulih yg mengarah kembali ke $x = 0$.

③. $\frac{dV}{dt} = f(t,V) = 9.81 - 0.08V^2$, $V(0) = 0$; $h = 0.5$

a. Euler

- $f(t_0, V_0) = 9.81 - 0.08(0)^2 = 9.81 - 0 = 9.81$

  $V_1 = 0 + 0.5(9.81) = 0 + 4.9050 = 4.9050$
- $f(t_1, V_1) = 9.81 - 0.08(4.9050)^2 = 9.81 - 0.08(24.0590) = 9.81 - 1.9247 = 7.8853$

  $V_2 = 4.9050 + 0.5(7.8853) = 4.9050 + 3.9426 = 8.8476$
- $f(t_2, V_2) = 9.8 - 0.08(8.8476)^2 = 9.81 - 6.2625 = 3.5975$

  $V_3 = 8.8476 + 0.5(3.5475) = 8.8476 + 1.7738 = 10.6214$
- $f(t_3, V_3) = 9.8 - 0.08(10.6219)^2 = 9.81 - 9.0251 = 0.7899$

  $V_4 = 10.6219 + 0.5(0.7899) = 10.6219 + 0.3929 = 11.0138$

No.

Date

n	tn	Vn	f(tn.Vn)
0	0	0	9.8100
1	0.5	4.9050	7.8853
2	1.0	8.8976	3.5975
3	1.5	10.6219	0.7899
4	2.0	11.0138	0.1056

b. Runge kutta orde 4

$k_1 = f(t_n, v_n)$; $k_2 = f(t_n + h/2, v_n + h/2\, k_1)$; $k_3 = f(t_n + h/2, v_n + h/2\, k_2)$

$k_4 = f(t_n + h, v_n + h k_3)$, $v_{n+1} = v_n + h/6\,(k_1 + 2k_2 + 2k_3 + k_4)$

- $k_1 = 9.81 - 0.08(0)^2 = 9.81$

$V_2 = 0 + 0.25(9.81) = 2.4525$

$k_2 = 9.81 - 0.08(2.4525)^2 = 9.328819$

$V_3 = 0 + 0.25(9.328819) = 2.332205$

$k_3 = 9.81 - 0.08(2.332205)^2 = 9.374866$

$V_4 = 0 + 0.5(9.374866) = 4.687433$

$k_4 = 9.81 - 0.08(4.687433)^2 = 8.052238$

$\Sigma k = 55.269608$

$V_1 = 0 + \frac{0.5}{6}(55.269608) = 4.605801$

- $k_1 = 9.81 - 0.08(4.605801)^2 = 8.112928$

$V_2 = 4605801 + 0.25(8.112928) = 6.634033$

$k_2 = 9.81 - 0.08(6.634033)^2 = 6.289169$

$V_3 = 4603801 + 0.25(6.289169) = 6.178093$

$k_3 = 9.81 - 0.08(6.178093)^2 = 6.756493$

$V_4 = 4605801 + 0.5(6.756493) = 7.984047$

$k_4 = 9.81 - 0.08(7.984047) = 4.710399$

$\Sigma k = 38.914652$

$V_2 = 4605801 + \frac{0.5}{6}(38.914652) = 7.848688$

- $k_1 = 9.81 - 0.08(7.848688)^2 = 4.881847$

$V_2 = 7.848688 + 0.25(4.881847) = 9.069150$

$k_2 = 9.81 - 0.08(9.069150)^2 = 3.230091$

$V_3 = 7.848688 + 0.25(3.230091) = 8.656199$

$k_3 = 9.81 - 0.08(8.656199)^2 = 3.815618$

$V_4 = 7.848688 + 0.5(3.815618) = 9.756497$

$k_4 = 9.81 - 0.08(9.756497)^2 = 2.194841$

$\Sigma k = 21.168027$

$V_2 = 7.898688 + \frac{0.5}{6}(21.168027) = 9.612691$

$k_1 = 9.81 - 0.08(9.612691)^2 = 2.417699$

$V_2 = 9.612691 + 0.25(2.917699) = 10.217119$

$k_2 = 9.81 - 0.08(10.217119)^2 = 1.458896$

$V_3 = 9.612691 + 0.25(1.458896) = 9.977902$

$k_3 = 9.81 - 0.08(9.977902)^2 = 1.846116$

$V_4 = 9.612691 + 0.25(1.846116) = 10.535748$

$k_4 = 9.81 - 0.08(10.535748)^2 = 0.929890$

$\Sigma k = 9.957459$

$V_4 = 9.612691 + \frac{0.5}{6}(9.957459) = 10.442479$

n	tn	Vn	k1	k2	k3	k4
0	0	0	9.81	9.320819	9.379866	8.052238
1	0.5	9.605801	8.112928	6.289169	6.756993	9.710309
2	1.0	7.898688	9.881897	3.230091	3.815618	2.194841
3	1.5	9.612691	2.917699	1.958894	1.89616	0.929890
4	2.0	10.442479	-	-	-	-

Perbandingan Hasil

Metode	V (2.0)
Euler	11.0138
Runge Kutta orde 4	10.4425